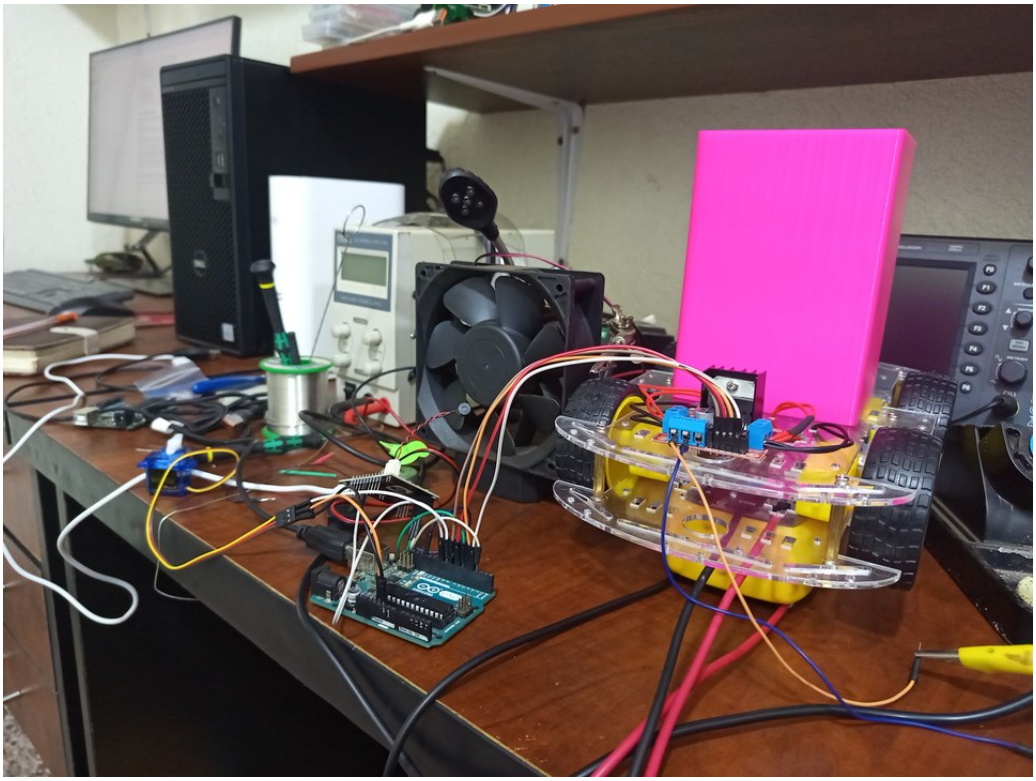
Smart Voice Controlled Dustbin

Hardware Setup Guide

Wiring, power, VC-02 configuration, assembly, and bring-up for the ATmega328P-based prototype.



Prototype integration on the 4-wheel chassis during bench testing.

Safety scope

This guide documents an educational prototype. It is not a certified medical device and must not be used for clinical waste without appropriate containment, cleaning, risk assessment, and regulatory approval.

1 System overview

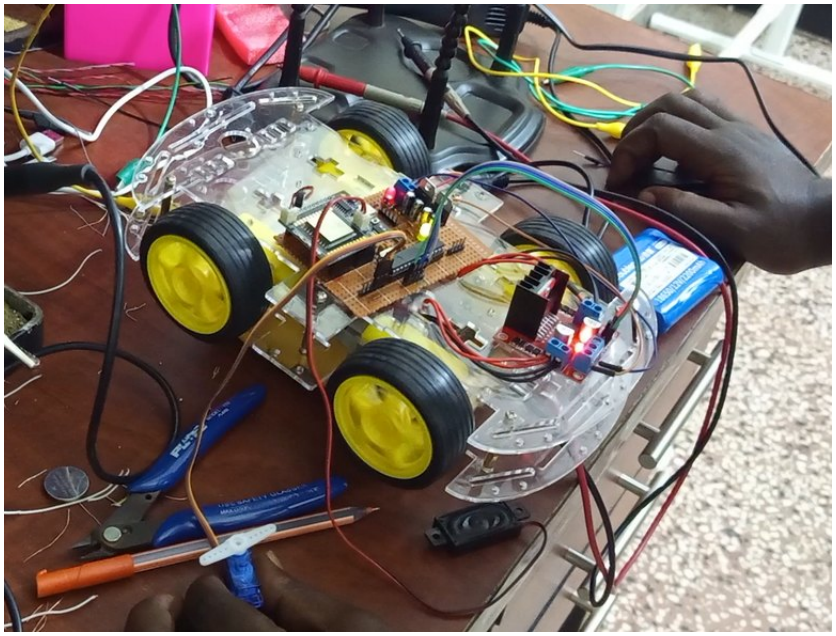
The VC-02 recognizes configured speech offline and exposes the result as GPIO pulses or levels. The ATmega328P reads those lines, drives two motor channels, monitors the ultrasonic sensor in the current direction of travel, moves a lid servo only while stopped, and reports status on an I2C LCD.

Implemented behavior

- Forward and backward commands are rising-edge pulse inputs.
- The active ultrasonic sensor stops motion at 20 cm or less.
- A maintained GPIO level opens the lid at HIGH and closes it at LOW.
- Motor PWM is fixed at 80 out of 255 in the supplied sketch.
- The system supports straight-line motion only; it does not steer or navigate autonomously.

Signal flow

Source	Signal	Destination	Effect
VC-02 GPIO_B0	200 ms pulse	ATmega D8	Start forward motion
VC-02 GPIO_B1	200 ms pulse	ATmega D7	Start backward motion
VC-02 GPIO_A27	HIGH / LOW	ATmega D3	Open / close lid
Ultrasonic sensors	Echo pulse	ATmega D1 / D2	Stop near obstacle
ATmega328P	PWM + direction	Motor driver	Drive the chassis



Early integration on perfboard and a dual-channel motor-driver module.

2 Parts and power

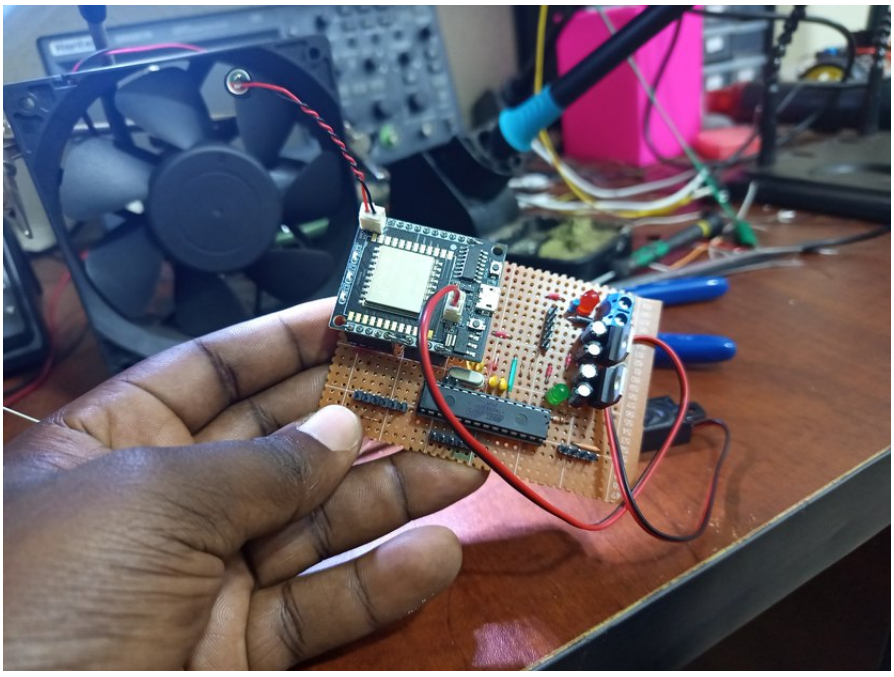
Core parts

Part	Role	Setup note
ATmega328P	Main controller	Use a complete 16 MHz circuit or an Arduino Uno-compatible board.
AI-Thinker VC-02	Offline voice recognition	Flash the configured command model before system testing.
Dual H-bridge driver	Motor direction and speed	Size it for motor stall current; the photographed module is L298N-style.
2 x ultrasonic sensor	Front and rear stopping	Mount square to the direction of travel with a clear acoustic path.
16x2 I2C LCD	Status display	Firmware expects address 0x27 on A4/A5.
Servo	Lid actuation	Use a separate regulated rail if the servo causes resets.
Battery and regulators	System power	Choose ratings from measured motor stall and logic current.

Power architecture

- Feed the motors through the motor driver from a supply matched to the motors.
- Power the ATmega328P and LCD from a clean regulated logic rail.
- Power the VC-02 only at the voltage required by its exact carrier/module revision; verify its datasheet and board markings.
- A separate regulated servo rail is recommended. Join its ground to the controller ground.
- Connect every signal-source ground together: controller, VC-02, sensors, servo supply, and motor driver.
- Add a fuse near the battery, a master switch, and local decoupling. Keep motor wiring away from voice and sensor wiring.

Do not power motors from the ATmega328P 5 V pin. Do not apply 5 V to a VC-02 GPIO unless that exact board is documented as 5 V tolerant.



Custom controller board with the VC-02 and ATmega328P during assembly.

3 Complete controller pin map

The mapping below is taken directly from `robot_controller.ino` and the supplied VC-02 pin note.

ATmega pin	Connected device	Device pin	Direction / function
D0 / RX	Rear ultrasonic	TRIG	Output; used during backward motion
D1 / TX	Rear ultrasonic	ECHO	Input; used during backward motion
D2	Front ultrasonic	ECHO	Input; used during forward motion
D3	VC-02	GPIO_A27	Input; HIGH opens, LOW closes
D5 (PWM)	Motor driver	ENA	Motor A speed
D6 (PWM)	Motor driver	ENB	Motor B speed
D7	VC-02	GPIO_B1	Input; backward pulse
D8	VC-02	GPIO_B0	Input; forward pulse
D9	Servo	Signal	Output; lid angle
D10	Front ultrasonic	TRIG	Output; used during forward motion
D11	Motor driver	IN1	Motor A direction
D12	Motor driver	IN2	Motor A direction
D13	Motor driver	IN3	Motor B direction
A0 / D14	Motor driver	IN4	Motor B direction
A4	I2C LCD	SDA	I2C data
A5	I2C LCD	SCL	I2C clock

Programming caution

D0 and D1 are the ATmega328P hardware UART pins. Their ultrasonic connections can interfere with bootloader uploads or serial debugging. Disconnect that sensor from D0/D1 while uploading if necessary. The current sketch does not use Serial, but future debugging will require relocating or isolating those signals.

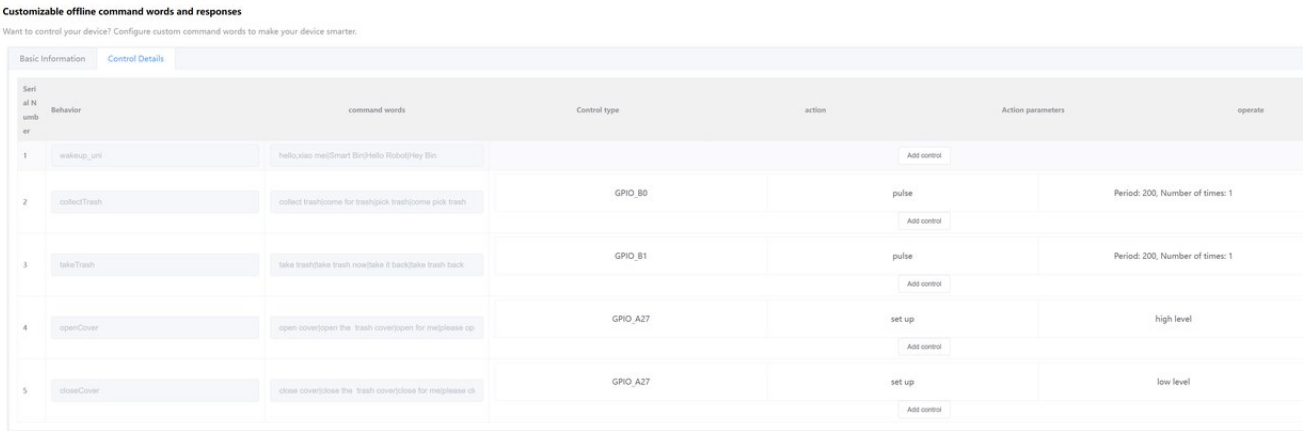
Motor outputs

Connect the left motor group to one H-bridge output and the right motor group to the other. If the platform rotates instead of traveling straight, reverse one channel's motor leads. Keep ENA and ENB jumpers removed when those pins are driven by D5 and D6 PWM.

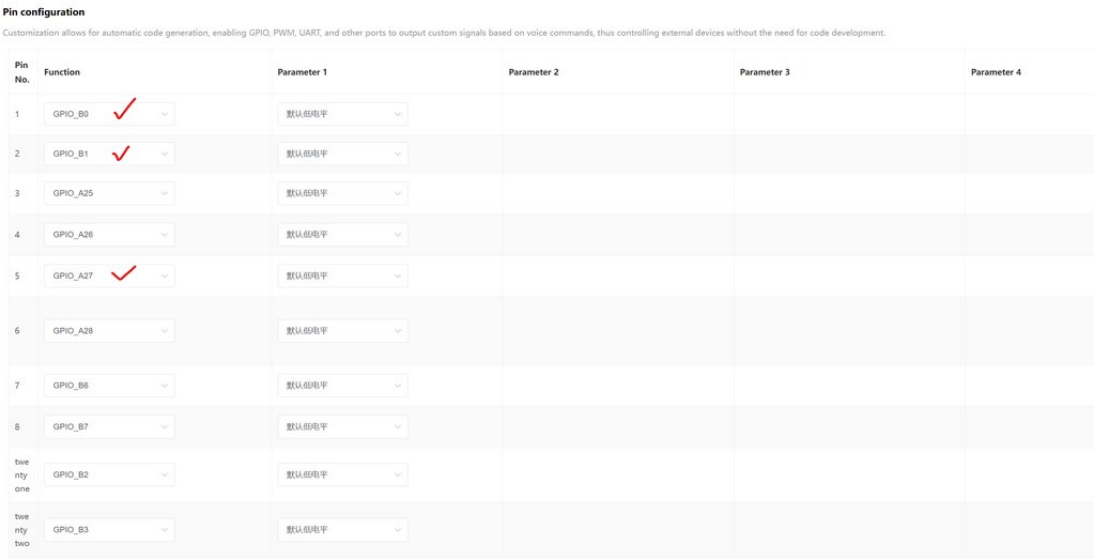
4 VC-02 configuration

Use the AI-Thinker voice platform linked in the repository note. Configure the three used outputs as shown below, generate the SDK/firmware, then flash the VC-02 using the vendor procedure for the exact module revision.

Behavior	Recognized phrase examples	Output setup
collectTrash	collect trash; come for trash; pick trash; come pick trash	GPIO_B0; pulse; 200 ms; once
takeTrash	take trash; take trash now; take it back; take trash back	GPIO_B1; pulse; 200 ms; once
openCover	open cover; open the trash cover; open for me; please open	GPIO_A27; set HIGH
closeCover	close cover; close the trash cover; close for me; please close	GPIO_A27; set LOW



Supplied control-detail screen showing pulse and level actions.



GPIO_B0, GPIO_B1, and GPIO_A27 selected in the voice-platform pin configuration.

5 Wiring and assembly sequence

1 Prepare the power system

With the battery disconnected, install the fuse and master switch. Verify the motor rail and regulated logic rails using a multimeter before connecting any module.

2 Wire the controller and display

Connect the ATmega328P support circuit and I2C LCD. Upload the sketch with D0/D1 loads disconnected. Confirm the display initializes at address 0x27 and shows the waiting message.

3 Connect the motor driver

Wire D5/D6 to ENA/ENB and D11/D12/D13/A0 to IN1-IN4. Connect each motor side to one output channel. Test with wheels lifted and use a current-limited supply.

4 Install ultrasonic sensors

Mount one sensor facing the rear and connect TRIG/ECHO to D0/D1. Mount the other facing forward and connect ECHO/TRIG to D2/D10. Power them from the logic rail and share ground.

5 Connect the servo

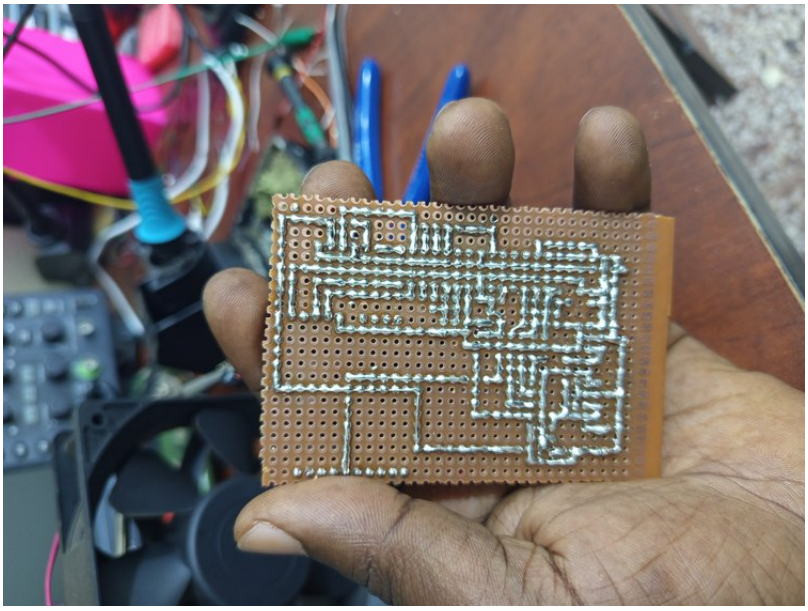
Connect the signal to D9. Before attaching the horn, power the controller once so the sketch commands 0 degrees; then fit the mechanism in the closed position without binding.

6 Connect the voice module

Connect GPIO_B0 to D8, GPIO_B1 to D7, and GPIO_A27 to D3. Add the common ground, then power the VC-02 from its verified rail. Keep voice wiring separated from the motor paths.

7 Secure and inspect

Strain-relieve battery and motor cables, insulate exposed conductors, check polarity, and confirm that no wiring can reach the wheels. The supplied STL enclosure can protect the custom control circuit after fit and temperature checks.



Perfboard solder side. Inspect for bridges, cold joints, and insufficient clearance before power-up.

6 Bring-up and verification

Use this order to limit mechanical and electrical risk.

Step	Test	Expected result
1	Continuity and polarity with power off	No short between rails; correct connector polarity
2	Logic-only power	Stable regulated voltage; LCD shows waiting status
3	Motor test with chassis raised	Both sides move forward and backward together
4	Front obstacle test	Forward motion stops at about 20 cm
5	Rear obstacle test	Backward motion stops at about 20 cm
6	Servo test while stopped	Smooth travel between 0 and 90 degrees
7	Voice GPIO test	Each configured phrase produces the intended action
8	Floor test at low risk	Straight travel; no resets, overheating, or cable contact

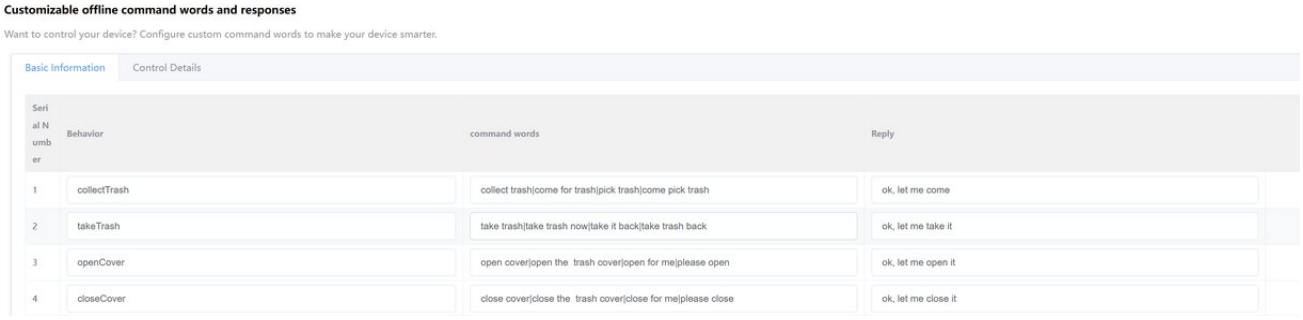
Fault checks

Symptom	Likely checks
Upload fails	Disconnect D0/D1 sensor; verify board, bootloader, serial adapter, and reset wiring.
LCD blank	Confirm 0x27 address, SDA=A4, SCL=A5, contrast setting, 5 V, and ground.
Robot turns	Reverse one motor channel or correct IN wiring; compare wheel direction with chassis raised.
Random resets	Measure rail sag; separate servo/motor supply; add decoupling; improve grounding.
Does not stop	Check TRIG/ECHO orientation and active sensor; test pulse timing and sensor field of view.
Voice action wrong	Confirm GPIO_B0->D8, GPIO_B1->D7, GPIO_A27->D3 and regenerate VC-02 firmware.

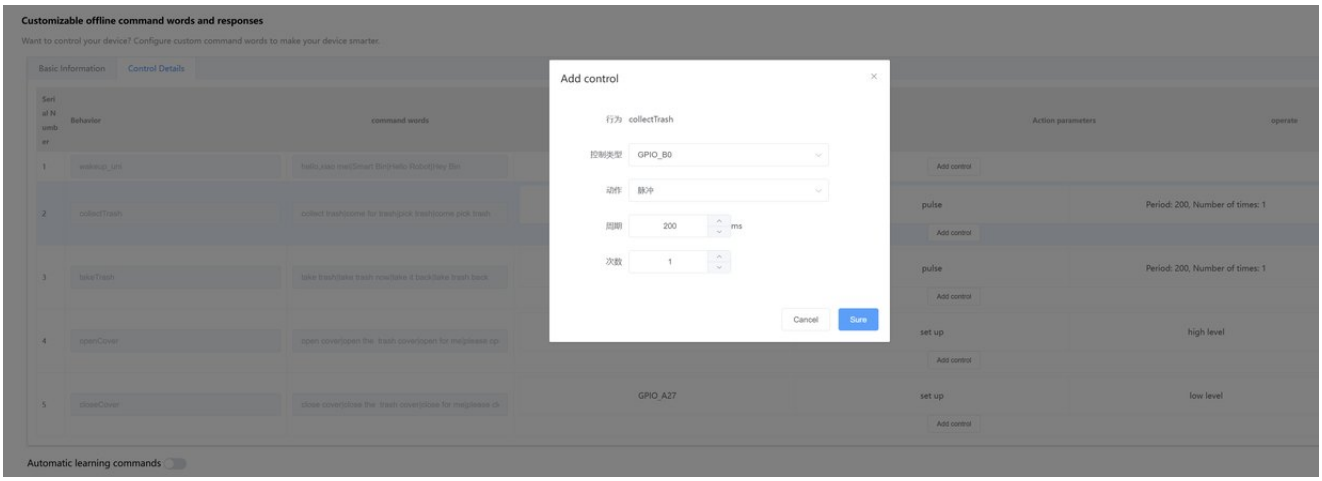
Fail-safe limitation

If an ultrasonic echo times out, the supplied firmware returns -1 and continues motion because -1 is not treated as an obstacle. Keep the emergency power switch accessible and consider changing sensor faults to a stop condition before unattended testing.

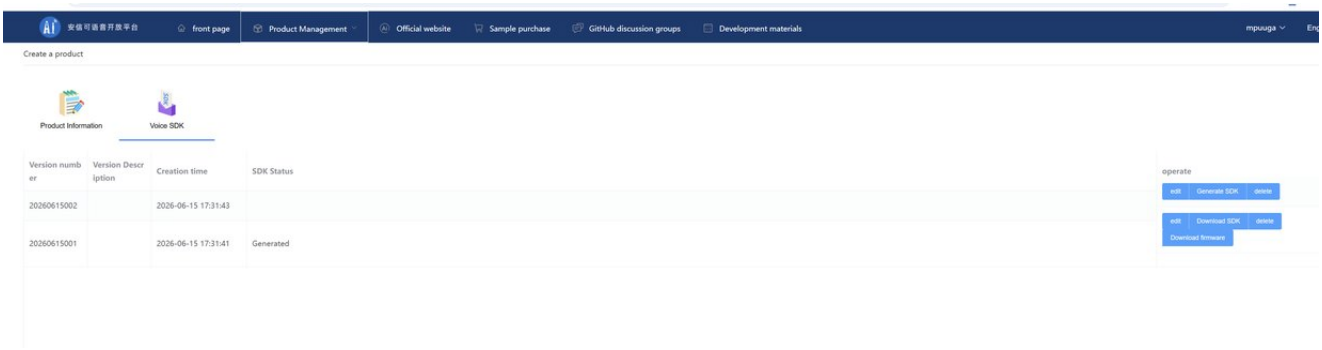
7 Configuration reference



Configured command vocabulary and spoken replies.



GPIO_B0 pulse action configured for the collectTrash behavior.



Voice-platform SDK generation and firmware download screen.

Reference links

AI-Thinker configuration platform: https://voice.ai-thinker.com/
Demonstration video: https://youtu.be/Hfe3AVP62qg

Source of truth

Firmware behavior and controller assignments are defined in `robot_controller/robot_controller.ino`. The VC-02-to-controller mapping is recorded in `info.txt`. Rebuild this PDF with `scripts/build_hardware_guide.py` after changing either source.